ECG Machine



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: ECG machine

What it is used for

An ECG machine records the electrical activity of the heart through electrodes attached to the patient's skin. Correct electrode placement, patient preparation, and signal quality checks help produce a reliable tracing for interpretation by qualified healthcare personnel.

Main parts

- ECG recorder and display
- Patient lead wires and electrode connectors
- Disposable or reusable electrodes
- Patient cable and lead set
- Control buttons and recording settings
- Power supply, battery, and printer or digital output

Before you start

1. Use the ECG machine only if trained and authorised to perform the procedure.
2. Confirm the patient's identity, explain the procedure, and obtain consent according to facility policy.
3. Check the machine, patient cable, lead wires, electrodes, power supply, and accessories for damage.
4. Verify that the machine is clean, functional, and within its required maintenance and calibration schedule.
5. Prepare a private, comfortable area and ensure the patient can lie still and relaxed.
6. Clean and dry electrode sites as required, and prepare the correct electrodes and lead configuration.

How to use

1. Position the patient comfortably, usually lying on their back, and ask them to relax and remain still.
2. Switch on the ECG machine and select the appropriate patient and recording settings.
3. Enter the required patient details accurately, following the facility's privacy procedures.
4. Attach electrodes to the correct anatomical locations using the approved lead-placement diagram.
5. Connect the lead wires to the corresponding electrodes and check that each connection is secure.
6. Ensure the patient cable is arranged to minimise pulling, movement, and electrical interference.

7. Check the displayed signal for a stable baseline and acceptable waveform quality before recording.
8. Ask the patient to remain still and breathe normally during acquisition unless otherwise instructed.
9. Start the recording and verify that all required leads and waveform segments have been captured.
10. Repeat the recording if movement artefact, loose electrodes, or signal interference makes the tracing unsuitable.
11. Label, save, or print the ECG with the correct patient identification, date, and time.
12. Provide the tracing to the authorised healthcare professional for interpretation; do not independently diagnose from the recording.
13. Disconnect the lead wires gently, remove the electrodes, and help the patient return to a comfortable position.
14. Clean reusable accessories and the machine's patient-contact surfaces according to infection-control procedures.

⚠ Safety

- ECG recording should be performed by trained healthcare personnel and interpreted by an appropriately qualified professional.
- Maintain patient privacy, dignity, and consent throughout the procedure.
- Inspect cables and electrical accessories before use; do not use equipment with exposed wires or damaged connectors.
- Keep the machine and electrical connections away from liquids and wet surfaces.
- Use single-use electrodes for one patient only and follow infection-control procedures for reusable accessories.
- Do not use the ECG procedure to delay emergency care; chest pain, severe breathlessness, fainting, or other urgent symptoms require immediate clinical assessment.
- Follow manufacturer instructions and facility procedures when using the machine with other connected medical equipment.

Care & storage

- Clean the machine and reusable patient-contact accessories after each patient according to approved instructions.
- Inspect lead wires, clips, connectors, and cables regularly for wear or damage.
- Check battery condition, paper supply, and printer function where applicable.
- Arrange preventive maintenance and electrical safety testing according to facility requirements.
- Store the machine in a clean, dry, secure location and report recurring signal or equipment faults.

Common mistakes to avoid

- Placing electrodes in incorrect anatomical positions.
- Recording with loose electrodes, dry contact surfaces, or poorly connected lead wires.
- Allowing patient movement or cable interference to degrade the tracing.

- Saving or printing a tracing with incorrect patient identification.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.